

THE BEST OPEN SOURCE SECURITY TOOLS

Security is an important part of an administrator's job. This article presents what the author believes are the 10 most useful security tools.



Nmap

Nmap is an open source tool created by Gordon Fyodor Lyon that supports port scanning, operating system detection, version detection and more. Nmap can be useful to both network administrators and hackers! This very versatile tool for network mapping lets you learn many things about your hosts and their status. You should experiment with it in order to improve your knowledge, and read its man page (*man nmap*) to learn all about its possibilities.

Network scanning is very useful for both checking and improving your network security. It's considered a good practice to periodically run Nmap and check for possible changes to its output.



Tcpdump and Wireshark

Tcpdump is a very capable command-line utility that allows you to capture network data. It is based on *libpcap*, which is an open source C/C++ library for network traffic capture. You can capture your traffic using *tcpdump* but you can analyse it later using Wireshark—another powerful tool that every Linux or network administrator should know about. *Tcpdump* has relatively simple, yet powerful commands; the

following *Tcpdump* command captures the traffic of the entire 10.10.10.0/24 network:

```
$ sudo tcpdump net 10.10.10.0/24
```

Wireshark also captures packets, and analyses and displays them in a human-readable format. Wireshark allows you to follow a TCP/IP 'conversation' between two machines, view the data of the packets, etc. I suggest that you start by first learning Display Filters.

The main advantage of Wireshark over *tcpdump* is that it is a graphical application—but there is also a command-line version of Wireshark. Remember that Wireshark has an overhead, as a GUI application, and can lose network data on a busy network; *Tcpdump* requires less system resources than Wireshark.

Before you start capturing, it is better to have in mind a particular problem that you want to solve or examine. This is the first step towards successful network traffic analysis.



Nessus security scanner

Nessus is a vulnerability scanning program that can scan for the following types of vulnerabilities: